

TASK 4 REPORT

Title

Model Fairness, Bias, and Explainability Analysis Using SHAP

Objective

The objective of this project is to analyze the fairness, bias, and explainability of a Machine Learning model using the Iris dataset. A Random Forest Classifier was trained, feature importance was analyzed, SHAP (SHapley Additive Explanations) was used to explain model predictions, and practical bias mitigation strategies were discussed.

Dataset

- **Dataset Name:** Iris Species Dataset
- **Source:** Kaggle / Scikit-learn Iris Dataset
- **Number of Features:** 4
- **Target Classes:** Setosa, Versicolor, Virginica

Features used:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Tools and Libraries

- Python
- Google Colab
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

- SHAP

Methodology

1. Loaded the Iris dataset.
2. Performed data preprocessing.
3. Split the dataset into training and testing sets.
4. Trained a Random Forest Classifier.
5. Evaluated model performance using Accuracy, Classification Report, and Confusion Matrix.
6. Calculated feature importance.
7. Used SHAP to explain model predictions globally and locally.
8. Discussed fairness analysis and bias mitigation techniques.

Results

- Random Forest model trained successfully.
- High classification accuracy was achieved.
- Feature Importance showed that **Petal Length** and **Petal Width** were the most influential features.
- SHAP Summary Plot explained the overall feature impact.
- SHAP Waterfall Plot explained an individual prediction.

Fairness Analysis

The Iris dataset contains only flower measurements and does not include sensitive attributes such as gender, race, or age. Therefore, fairness analysis across protected groups could not be performed. In real-world applications, fairness metrics should be evaluated using datasets containing protected attributes.

Bias Mitigation Recommendations

- Use balanced datasets during model training.

- Carefully evaluate sensitive attributes if present.
- Monitor fairness metrics regularly.
- Use explainable AI techniques such as SHAP or LIME.
- Retrain the model if significant bias is detected.

Conclusion

The Random Forest model successfully classified Iris flower species with high accuracy. SHAP explainability improved model transparency by showing how individual features influenced predictions. Although fairness analysis was not applicable due to the absence of protected attributes in the dataset, appropriate bias mitigation strategies were identified. This project demonstrates the importance of explainable and fair machine learning models.

Outcome

- Successfully trained and evaluated a Random Forest model.
- Generated Feature Importance visualization.
- Performed SHAP-based model explainability.
- Completed fairness analysis.
- Proposed bias mitigation strategies.
- Successfully fulfilled the objectives of Task 4.

TASK4_INTERNSPARK.ipynb

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More actions

model.fit(X_train, y_train)

RandomForestClassifier

RandomForestClassifier(random_state=42)

predictions = model.predict(X_test)

print(predictions)

[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 2 2 2 0 0]

from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)

Accuracy: 1.0

from sklearn.metrics import classification_report

print(classification_report(y_test, predictions))

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

from sklearn.metrics import confusion_matrix

Variables Terminal

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import matplotlib.pyplot as plt

importance = model.feature_importances_

plt.figure(figsize=(8,5))

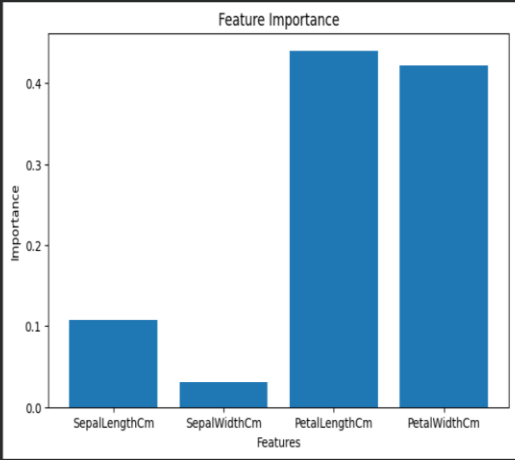
plt.bar(X.columns, importance)

plt.title("Feature Importance")

plt.xlabel("Features")

plt.ylabel("Importance")

plt.show()



Features	Importance
SepalLengthCm	0.12
SepalWidthCm	0.05
PetalLengthCm	0.45
PetalWidthCm	0.45

Variables Terminal

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